Q1 2021 (01 Sep Shift 2)

Which of the following is equivalent to the Boolean expression $p \wedge \sim q$?

(1) $\sim(q \rightarrow p)$

(2) $\sim p \rightarrow \sim q$

(3) $\sim(p \rightarrow \sim q)$

(4) $\sim(p \rightarrow q)$

Q2 2021 (31 Aug Shift 2)

Negation of the statement $(p \vee r) \Rightarrow (q \vee r)$ is :

(1) $p \wedge \sim q \wedge \sim r$

(2) $\sim p \wedge q \wedge \sim r$

(3) $\sim p \wedge q \wedge r$

(4) $p \wedge q \wedge r$

Q3 2021 (31 Aug Shift 1)

Let $*, \square \in \{\wedge, \vee\}$ be such that the Boolean expression $(p * \sim q) \Rightarrow (p \square q)$ is a tautology. Then :

(1) $* = \vee, \square = \vee$

(2) $* = \wedge, \square = \wedge$

(3) $* = \wedge, \square = \vee$

(4) $* = \vee, \square = \wedge$

Q4 2021 (27 Aug Shift 2)

The Boolean expression $(p \wedge q) \Rightarrow ((r \wedge q) \wedge p)$ is equivalent to:

(1) $(p \wedge q) \Rightarrow (r \wedge q)$

(2) $(q \wedge r) \Rightarrow (p \wedge q)$

(3) $(p \wedge q) \Rightarrow (r \vee q)$

(4) $(p \wedge r) \Rightarrow (p \wedge q)$

Q5 2021 (27 Aug Shift 1)

The statement $(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$ is :

(1) a tautology

(2) equivalent to $p \rightarrow \sim r$

(3) a fallacy

(4) equivalent to $q \rightarrow \sim r$

Q6 2021 (26 Aug Shift 2)

Consider the two statements:

$(S1)$: $(p \rightarrow q) \vee (\sim q \rightarrow p)$ is a tautology.

$(S2)$: $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a fallacy.

Then :

(1) only (S1) is true.

(2) both (S1) and (S2) are false.

(3) both (S1) and (S2) are true.

(4) only (S2) is true.

Q7 2021 (26 Aug Shift 1)

If the truth value of the Boolean expression $((p \vee q) \wedge (q \rightarrow r) \wedge (\sim r)) \rightarrow (p \wedge q)$ is false then the truth values of the statements p, q, r respectively can be:

(1) TFT

(2) FFT

(3) TFF

(4) FTF

Answer Key

Q1 (4)	**Q2** (1)	**Q3** (3)	**Q4** (1)
Q5 (1)	**Q6** (3)	**Q7** (3)	

Q1 (4)

p	q	$\sim p$	$\sim q$	$p - q$	$\sim(p \rightarrow q)$	$q \rightarrow p$	$\sim(q \rightarrow p)$
T	T	F	F	T	F	T	F
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	F	T	F

$p \wedge \sim q$	$\sim p \rightarrow \sim q$	$p \rightarrow \sim q$	$\sim(p \rightarrow \sim q)$
F	T	F	T
T	T	T	F
F	F	T	F
F	T	T	F

$p \wedge \sim q \equiv \sim (p \rightarrow q)$ Option (4)

Q2 (1)

$\because \sim (A \Rightarrow B) = A \wedge \sim B$

$\therefore \sim ((p \vee r) \Rightarrow (q \vee r))$

$= (p \vee r) \wedge (\sim q \wedge \sim r)$

$= ((p \vee r) \wedge (\sim r)) \wedge (\sim q)$

$= p \wedge (\sim r) \wedge (\sim q)$

Q3 (3)

$(p \wedge \sim q) \rightarrow (p \vee q)$ is tautology

p	q	$\sim q$	$p \wedge \sim q$	$p \vee q$	$(p \wedge \sim q) \rightarrow (p \vee q)$
T	T	F	F	T	T
T	F	T	T	T	T
F	T	F	F	T	T
F	F	T	F	F	T

Q4 (1)

$(p \wedge q) \Rightarrow ((r \wedge q) \wedge p)$

$\sim (p \wedge q) \vee ((r \wedge q) \wedge p)$

$\sim (p \wedge q) \vee ((r \wedge p) \wedge (p \wedge q)$

$\Rightarrow [\sim (p \wedge q) \vee (p \wedge q)] \wedge (\sim (p \wedge q) \vee (r \wedge p))$

$\Rightarrow t \wedge [\sim (p \wedge q) \vee (r \wedge p)]$

$\Rightarrow \sim (p \wedge q) \vee (r \wedge p)$

$\Rightarrow (p \wedge q) \Rightarrow (r \wedge p)$

Aliter :

given statement says

" if p and q both happen then

p and q and r will happen"

it Simply implies

" If p and q both happen then

'r' too will happen "

i.e.

" if p and q both happen then r and p too will happen

i.e.

$(p \wedge q) \Rightarrow (r \wedge p)$

Q5 (1)

$(\mathrm{p} \wedge (\mathrm{p} \rightarrow \mathrm{q}) \wedge (\mathrm{q} \rightarrow \mathrm{r})) \rightarrow \mathrm{r}$

$\equiv (\mathrm{p} \wedge (\sim \mathrm{p} \vee \mathrm{q}) \vee (\sim \mathrm{q} \vee \mathrm{r})) \rightarrow \mathrm{r}$

$\equiv ((\mathrm{p} \wedge \mathrm{q}) \wedge (\sim \mathrm{p} \vee \mathrm{r})) \rightarrow \mathrm{r}$

$\equiv (\mathrm{p} \wedge \mathrm{q} \wedge \mathrm{r}) \rightarrow \mathrm{r}$

$\equiv \sim (\mathrm{p} \wedge \mathrm{q} \wedge \mathrm{r}) \vee \mathrm{r}$

$\equiv (\sim \mathrm{p}) \vee (\sim \mathrm{q}) \vee (\sim \mathrm{r}) \vee \mathrm{r}$

$\Rightarrow$ tautology

Q6 (3)

$S_1 : (\sim p \vee q) \vee (q \vee p) = (q \vee \sim p) \vee (q \vee p)$

$S_1 = q \vee (\sim p \vee p) = qvt = t = \text{tautology}$

$S_2 : (p \wedge \sim q) \wedge (\sim p \vee q) = (p \wedge \sim q) \wedge \sim (p \wedge \sim q) = C$

$= \text{fallacy}$

Q7 (3)

p	q	r	$\underbrace{p \vee q}_{a}$	$\underbrace{q \to r}_{b}$	$a \wedge b$	$\sim r$	$\underbrace{a \wedge b \wedge (\sim r)}_{c}$	$\underbrace{p \wedge q}_{d}$	$c \to d$
T	F	T	T	T	T	F	F	F	T
F	F	T	F	T	F	F	F	F	T
T	F	F	T	T	T	T	T	F	F
F	T	F	T	F	F	T	F	F	T